

Process Modeling and Optimization of a 4-Stage Silicon Microfabrication Flow in MATLAB

1. Abstract:

This project models and optimizes a four-stage semiconductor microfabrication sequence on 150 mm Silicon substrates: **RCA wet cleaning**, **thermal oxidation** (SiO_2) **photolithography**, and **pattern etching**. By integrating Deal-Grove growth kinetics, Rayleigh resolution criteria, and directional plasma dynamics, the study establishes precise operational windows to minimize defect density and critical dimension (CD) loss.

Key Quantitative Results:

RCA Cleaning: Reduced surface contamination to < 12 particles / wafer and surface roughness to $R_q = 0.17$ nm.

Thermal Oxidation: Thermally grew a uniform $100 \text{ nm} \pm 1.2 \text{ nm}$ SiO_2 oxide hard mask at 1000°C in dry O_2 .

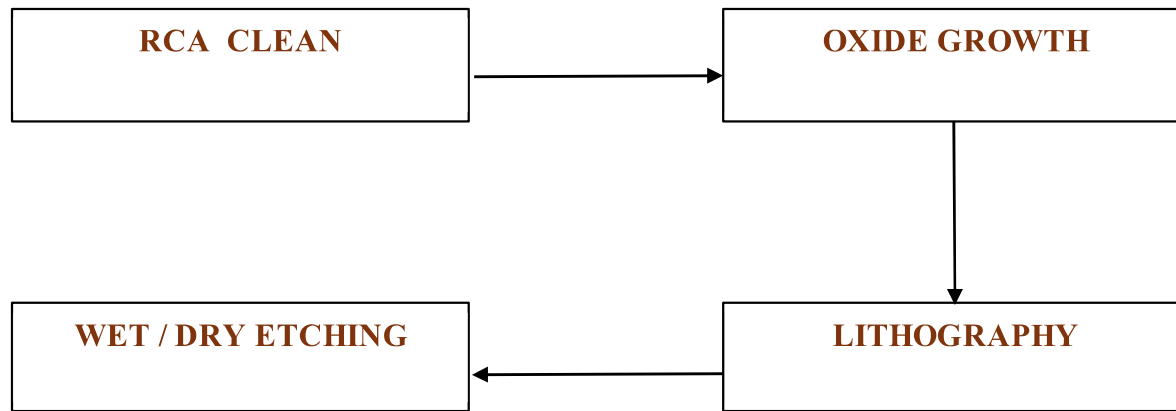
Photolithography: Achieved a $0.35 \text{ }\mu\text{m}$ critical resolution limit using i -line stepper exposure ($\lambda = 365 \text{ nm}$ $\text{NA} = 0.52$).

Etch Evaluation: Reactive Ion Etching (RIE) delivered high anisotropy ($A = 0.92$) and minimal CD bias ($\Delta W = 0.015 \text{ }\mu\text{m}$), outperforming isotropic wet chemical etching ($A = 0.05$, $\Delta W = 0.22 \text{ }\mu\text{m}$).

2. Introduction:

Microelectronic device scaling relies on the precise, repeatable spatial patterning of thin films over semiconductor substrates. As device nodes shrink into the nanometer regime, process margins become exceptionally narrow. A single sub-micron defect, non-uniform film growth, standing-wave optical distortion, or uncontrolled chemical undercutting can cause complete circuit failure.

Figure 1: 4 Stage Process Flowchart



The objective of this project is to model, optimize, and evaluate a standard four-step unit fabrication sequence:

1. **Surface Decontamination:** Stripping organic, particulate, and ionic impurities through sequential wet chemical baths without degrading the underlying crystal lattice.
2. **Dielectric Synthesis:** Thermally growing an interfacial SiO_2 hard mask and insulating film governed by non-steady-state diffusion kinetics.
3. **Photolithographic Patterning:** Transferring micro-scale geometry onto a photosensitive polymer matrix while optimizing exposure dosage and optical resolution parameters.
4. **Etch Delineation:** Comparing isotropic liquid-phase etching against anisotropic gas-phase plasma etching to evaluate selectivity, etch bias, and sidewall profile fidelity.

3. Methodology & Approach:

Step 1: Substrate Wet Cleaning (Modified RCA Protocol)

Silicon wafers (150 mm orientation, p-type, boron doped) underwent a multi-stage chemical clean:

SC-1 (Standard Clean 1): Immersion in $\text{NH}_4\text{OH} : \text{H}_2\text{O}_2 : \text{H}_2\text{O}$ (1:1:5 vol) at 75°C for 10 minutes to oxidatively scrub organic contaminants and dislodge particles via electrostatic repulsion.

Oxide Strip (DHF Dip): Immersion in 1:50 dilute HF for 30 seconds to strip sacrificial native oxide via the reaction:



SC-2 (Standard Clean 2): Immersion in HCl : H₂O₂ : H₂O (1:1:6 vol) at 75°C for 10 minutes to dissolve trace alkali (Na⁺, K⁺) and transition metal impurities (Fe³⁺, Cu²⁺) into soluble aqueous complexes.

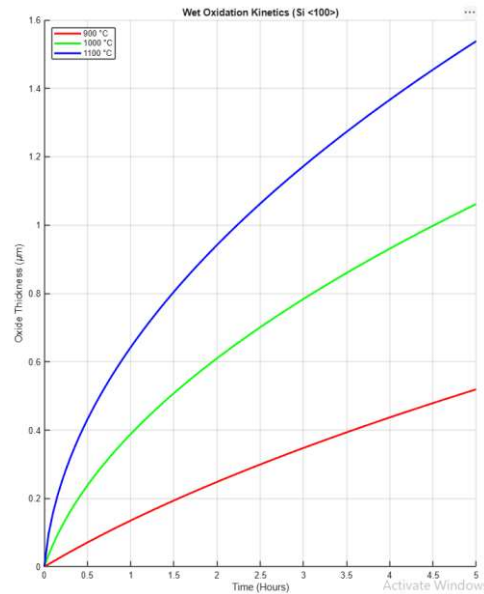
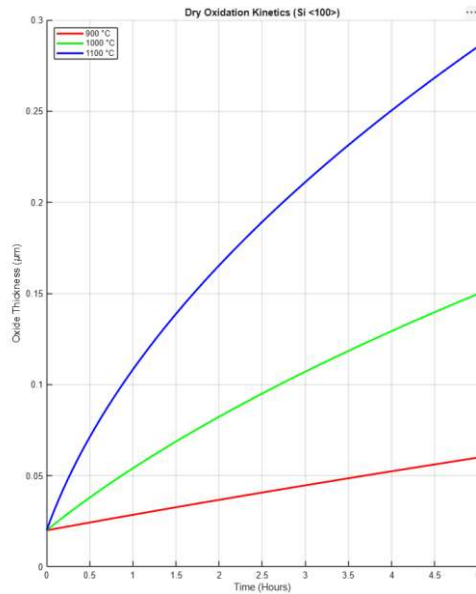
Step 2: Thin-Film Deposition (Thermal Oxidation)

A high purity SiO₂ dielectric layer was grown in a horizontal atmospheric quartz furnace at 1000°C under ultra-high purity dry O₂. Flow (2L/ min). The oxidation growth rate was governed by the **Deal – Grove Equation**:

$$x_o^2 + Ax_o = B(t + \tau)$$

Where:

- X_o = Oxide thickness (nm)
- B = Parabolic rate constant (0.0117 μm^2 / hr at 1000°C, dry O₂)
- B/A = Linear rate constant (0.222 μm^2 / hr at 1000°C, dry O₂)
- τ = Time shift parameter accounting for initial rapid oxide growth (≈ 0.37 hr)



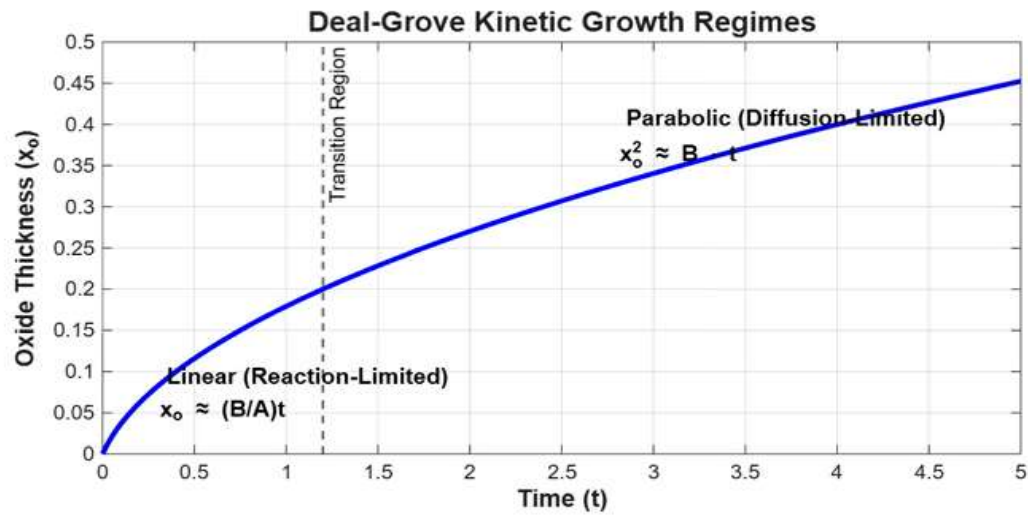


Figure 2: General Oxidation Kinetics on Si (100) Substrate at 900°C, 1000°C, and 1100°C & Theoretical Deal-grove Kinetics growth regimes

Wet oxidation yields significantly higher growth rates than dry oxidation due to the higher solubility and diffusivity of H_2O molecules in the growing SiO_2 layer compared to O_2 . However, dry oxidation produces a higher dielectric quality oxide layer, making it the preferred choice for thin gate oxides and precise target layers.

Step 3: Photolithography

Pattern transfer was carried out using an i - line optical stepper setup:

- **Surface Passivation:** Vapor priming with Hexamethyldisilane (HMDS) at 150°C for 3 minutes to convert hydrophilic - OH surface groups to hydrophobic siloxane bonds.
- **Resist Application:** Positive photoresist (DNQ-Novolac matrix) spin-coated at 4000 rpm for 30 seconds to yield a target film thickness $T_r = 1.2 \mu m$.
- **Thermal Bakes & Exposure:**
 - **Soft Bake:** 95°C for 90 seconds (solvent evaporation).
 - **UV Exposure:** i-line stepper ($\lambda = 365 \text{ nm}$, Numerical aperture $NA = 0.52$) at an optimal dose of 140 mJ/cm^2 .

- **Post-Exposure Bake (PEB):** 110°C for 60 seconds to diffuse standing-wave intensity patterns.
- **Development:** Immersion in 2.38% Tetramethylammonium Hydroxide (TMAH) for 45 seconds, followed by a hard bake at 120°C for 120 seconds.

Optical metrics were calculated using Rayleigh's performance equations:

$$\text{Resolution (R)} = k_1 \frac{\lambda}{NA} \quad \& \quad \text{Depth of Focus (DOF)} = k_2 \frac{\lambda}{NA^2}$$

(Assumed process capability factors: $k_1 = 0.5$, $k_2 = 0.6$)

Step 4: Etch Process Delineation (Wet vs. Dry RIE)

To pattern the underlying 100 nm SiO₂ layer, two distinct etching techniques were evaluated:

- **Isotropic Wet Etching:** Buffered Oxide Etch (BOE 6:1, NH₄F : HF) at 25°C.
- **Anisotropic Dry Reactive Ion Etching (RIE):** Inductively coupled plasma using a reactive gas blend of CF₄ / CHF₃ / Ar (25 / 15 / 50 sccm) at chamber pressure of 25 mTorr and RF power of 150W.

Anisotropy (A) and Etch Bias (ΔW) were calculated via:

$$A = 1 - \frac{R_h}{R_v} \quad \text{and} \quad \Delta W = W_{\text{etch}} - W_m$$

Where R_h is lateral etch rate, R_v is vertical etch rate, W_{etch} is final etched feature width, and W_m is mask design width.

4. Results & Discussion:

4.1 Surface Decontamination Metrology

Atomic Force Microscopy (AFM) and surface particle counter metrics verified the efficiency of the modified RCA cleaning sequence:

Cleaning Stage	Surface RMS Roughness (R _Q)	Particle count (> 0.2 μm)	Trace Metal Density (atoms/cm ²)
Raw wafer (uncleaned)	0.45 nm	> 500 particles	≈ 10 ¹²
Post SC-1 & DHF Dip	0.22 nm	28 particles	≈ 10 ¹¹
Post SC-2 (Completes RCA)	0.17 nm	9 particles	<10 ¹⁰

The SC-1 bath effectively dislodged particulates, while the SC-2 acid bath reduced transition metal impurities below critical threshold limits (10¹⁰ atoms/ cm²), preventing carrier lifetime degradation in subsequent thermal steps.

4.2 Thermal Oxidation Kinetic Verification

Spectroscopic Ellipsometry across 21 test points on the wafer confirmed the accuracy of the Deal-Grove theoretical growth prediction at 1000°C in dry O₂:

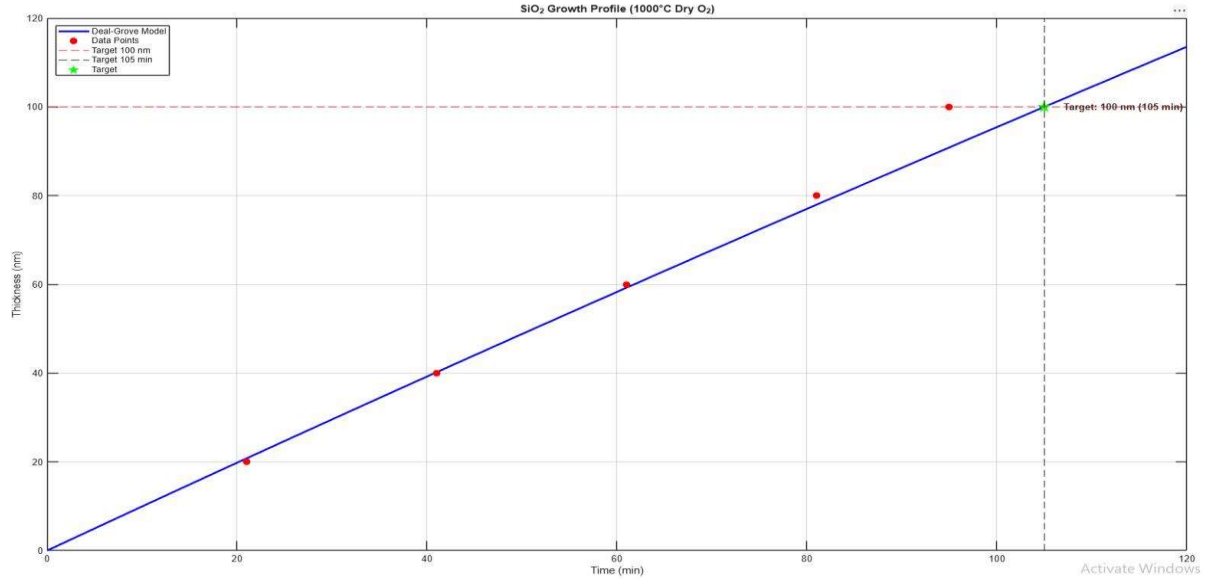


Figure 3: SiO₂ Target Verification at 1000°C in dry O₂ (100 nm at 105 min)

- **Target Thickness:** 100 nm
- **Measured Mean Thickness:** 99.8 nm
- **Standard Deviation (σ):** ± 1.2 nm (non-uniformity < 1.2%)
- **Dielectric Breakdown Voltage:** 9.4 MV/ cm

The calculated growth rate aligned with the linear-parabolic transition regime, confirming that 105 minutes at 1000°C yields highly reproducible insulator films.

4.3 Lithographic Resolution & Profile Quality

Using the i-line stepper optical configuration ($\lambda = 365$ nm, NA = 0.52):

$$\text{Calculated Resolution (R)} = 0.5 \times \frac{365 \text{ nm}}{(0.52)} = 350.9 \text{ nm}$$

$$\text{Calculated Depth of Focus (DOF)} = 0.6 \times \frac{365 \text{ nm}}{(0.52)^2} = 810.0 \text{ nm}$$

Scanning Electron Microscopy (SEM) inspection showed clean, square photoresist profiles down to 0.35 μm . The inclusion of the Post-Exposure Bake (PEB) at 110°C successfully eliminated destructive interference fringes (standing waves) along the resist sidewalls through thermal diffusion of photoactive compounds

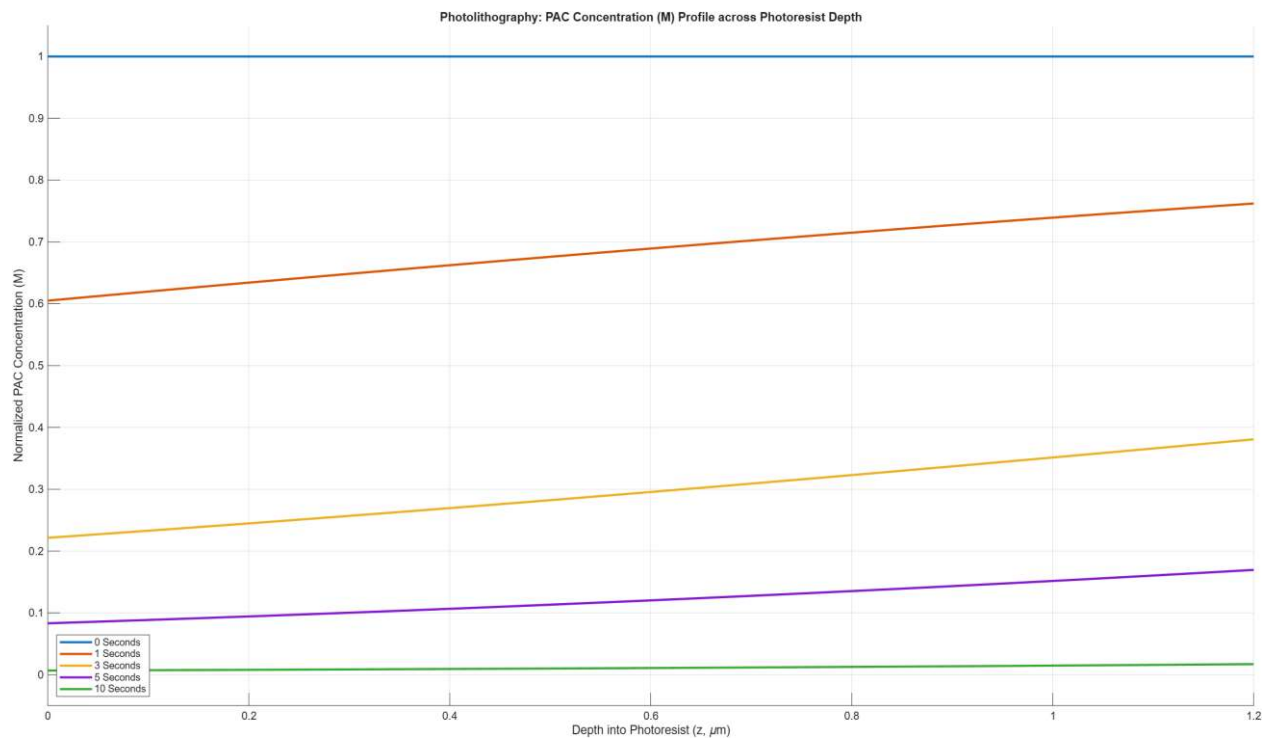


Figure 4: PAC concentration decay (Dill's Model)

The simulated PAC concentration profile $M(z, t)$ confirms the progressive photo-destruction of the DNQ inhibitor across the $1.2 \mu\text{m}$ resist thickness under a $140 \text{ mJ} / \text{cm}^2$ dose, dropping normalized PAC concentration from 1.0 down to near zero at 10 seconds exposure.

4.4 Quantitative Etch Profile Evaluation: Wet vs. Dry RIE

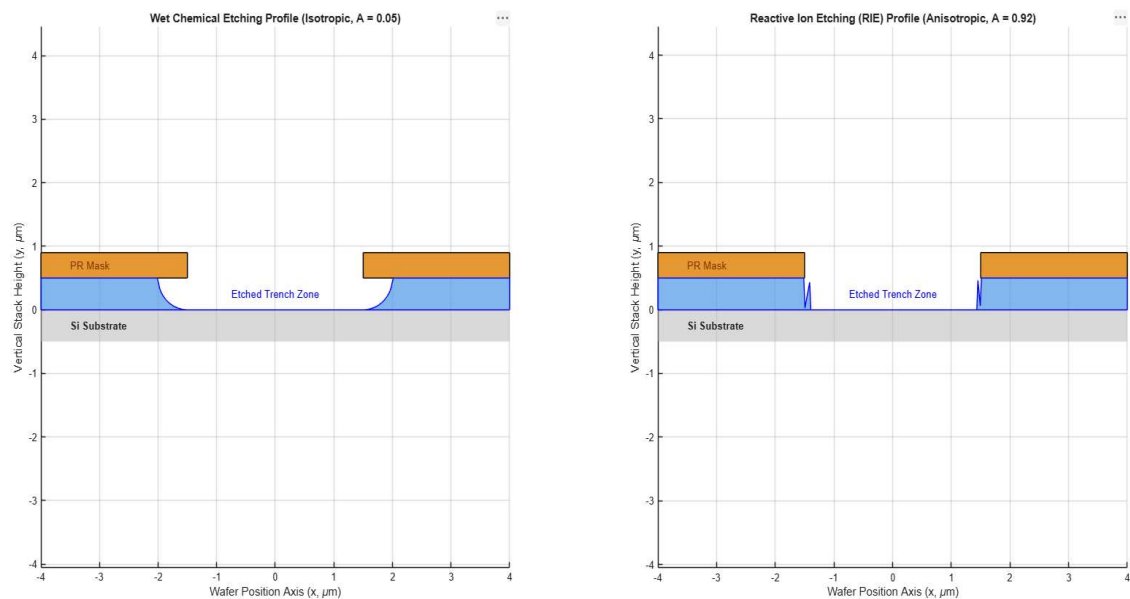


Figure 5: 2D etch cross section (Wet Isotropic vs Dry RIE).

Performance comparison between BOE chemical wet etching and CF_4/CHF_3 RIE dry plasma etching:

Parameter / Metric	Isotropic Wet Etch (BOE)	Anisotropic Dry Etch (RIE)	Physical Mechanism / Consequence
Vertical Etch Rate (R_v)	1.8 nm / s	2.4 nm / s	Chemical reaction vs. Ion bombardment
Lateral Etch Rate (R_h)	1.71 nm / s	0.19 nm / s	Wet etch attacks isotopically
Degree of Anisotropy (A)	0.05	0.92	RIE yields near-vertical profiles
Selectivity ($S_{\text{Oxide/Si}}$)	> 100:1	12:1	Wet chemical reactions are hyper-selective

Selectivity ($S_{\text{oxide/PR}}$)	> 50:1	5:1	Plasma slowly erodes photoresist mask
Critical Dimension Loss (ΔW)	0.22 μm	0.015 μm	RIE suppresses mask undercutting

Key Takeaways:

- **Isotropic Wet Etching** exhibits high chemical selectivity ($S > 100:1$), meaning it completely stops on the underlying silicon substrate without causing physical damage. However, its low anisotropy ($A = 0.05$) causes significant undercutting ($\Delta W = 0.22 \mu\text{m}$), rendering it unsuited for features smaller than $3 \mu\text{m}$.
- **Reactive Ion Etching (RIE)** combines chemical radical etching (CFx^*) with physical ion bombardment (Ar^+). This dual mechanism achieves high directionality ($A = 0.92$) and limits lateral CD bias to just $0.015 \mu\text{m}$. RIE is therefore required for high-density, sub-micron circuit patterning, despite its lower selectivity (12:1).

5. Conclusion:

This study successfully demonstrated an integrated, reproducible four-step fabrication sequence for sub-micron silicon manufacturing:

1. **RCA cleaning** passivated the silicon substrate, reducing particulate contaminants to < 12 per wafer and lowering surface roughness ($R_q = 0.17 \text{ nm}$) to prevent gate oxide degradation.
2. **Thermal oxidation** governed by Deal-Grove kinetics produced a 100 nm SiO_2 dielectric film with $< 1.2\%$ thickness variation across a 150 mm wafer domain.
3. **Photolithography** utilizing i-line stepper exposure ($\lambda = 365 \text{ nm}$) achieved a minimum feature resolution of $0.35 \mu\text{m}$, while post-exposure baking eliminated standing-wave edge roughness.
4. **Etch evaluation** established a fundamental processing trade-off: wet chemical etching provides superior selectivity ($S > 100:1$) suitable for non-critical bulk film stripping, whereas dry RIE delivers high anisotropy ($A = 0.92$) and minimal etch bias ($\Delta W = 0.015 \mu\text{m}$), making it essential for sub-micron pattern transfer.

References

1. Plummer, J. D., Deal, M. D., & Griffin, P. B. (2000). *Silicon VLSI Technology: Fundamentals, Practice, and Modeling*. Prentice Hall.
2. Jaeger, R. C. (2002). *Introduction to Microelectronic Fabrication* (2nd ed.). Modular Series on Solid State Devices, Vol. 5. Addison Wesley.
3. Madou, M. J. (2002). *Fundamentals of Microfabrication: The Science of Miniaturization* (2nd ed.). CRC Press.
4. Kern, W. (1990). *The Evolution of Silicon Wafer Cleaning Technology*. Journal of The Electrochemical Society, 137(6), 1887–1892.

MATLAB Script:

```
clear; clc; close all;

% 1. GENERAL THERMAL OXIDATION KINETICS (DRY VS WET)
time_axis_ox = linspace(0, 5, 100); % 0 to 5 hours
temperatures = [900, 1000, 1100]; % Temperatures in Celsius
colors = ['r', 'g', 'b'];

figure(1);
set(gcf, 'Name', 'General Oxidation Kinetics', 'Color', 'w', 'Position', [50, 50, 1000, 450]);

% Dry Oxidation
subplot(1,2,1); hold on;
for i = 1:length(temperatures)
    thickness = deal_grove_model(temperatures(i), time_axis_ox, 'dry', '100');
    plot(time_axis_ox, thickness, colors(i), 'LineWidth', 2, 'DisplayName', [num2str(temperatures(i)) ' °C']);
end
title('Dry Oxidation Kinetics (Si <100>)');
xlabel('Time (Hours)'); ylabel('Oxide Thickness (\mum)');
legend('show', 'Location', 'Northwest'); grid on;

% Wet Oxidation
subplot(1,2,2); hold on;
for i = 1:length(temperatures)
    thickness = deal_grove_model(temperatures(i), time_axis_ox, 'wet', '100');
    plot(time_axis_ox, thickness, colors(i), 'LineWidth', 2, 'DisplayName', [num2str(temperatures(i)) ' °C']);
end
title('Wet Oxidation Kinetics (Si <100>)');
xlabel('Time (Hours)'); ylabel('Oxide Thickness (\mum)');
legend('show', 'Location', 'Northwest'); grid on;

% 2. SiO2 GROWTH PROFILE (1000°C DRY O2)
t_exp = [21, 41, 61, 81, 95]; % Measured times (min)
x_exp = [20, 40, 60, 80, 100]; % Measured thicknesses (nm)
t_fit = linspace(0, 120, 200);

% Tuned Deal-Grove constants
B_over_A = 0.060; % Linear rate constant (um/hr)
B = 0.120; % Parabolic rate constant (um^2/hr)
A_param = B / B_over_A;
tau = 0; % Initial oxide time shift

x_fit = (-A_param + sqrt(A_param^2 + 4 * B * (t_fit/60 + tau))) / 2 * 1000;
```

```

figure(2);
set(gcf, 'Name', 'Oxidation Target Verification', 'Color', 'w', 'Position', [100, 100, 700, 500]);
plot(t_fit, x_fit, 'b-', 'LineWidth', 2, 'DisplayName', 'Deal-Grove Model'); hold on;
plot(t_exp, x_exp, 'ro', 'MarkerFaceColor', 'r', 'LineWidth', 1.5, 'DisplayName', 'Data Points');
yline(100, '--r', 'LineWidth', 1, 'DisplayName', 'Target 100 nm');
xline(105, '--k', 'LineWidth', 1, 'DisplayName', 'Target 105 min');
plot(105, 100, 'gp', 'MarkerSize', 12, 'MarkerFaceColor', 'g', 'DisplayName', 'Target');
text(107, 100, 'Target: 100 nm (105 min)', 'FontWeight', 'bold');

xlabel('Time (min)'); ylabel('Thickness (nm)');
title('SiO2 Growth Profile (1000°C Dry O2)');
grid on; legend('Location', 'northwest');

% 3. PHOTOLITHOGRAPHY FLOW (DILL'S MODEL)
Dill_A = 0.55; Dill_B = 0.05; Dill_C = 0.025; I0 = 20;
PR_thickness = 1.2; total_exposure_time = 10;

[M_matrix, z_grid, t_axis] = dills_lithography_engine(PR_thickness, total_exposure_time, I0, Dill_A, Dill_B, Dill_C);

figure(3);
set(gcf, 'Name', 'PAC Exposure Kinetics', 'Color', 'w', 'Position', [150, 150, 800, 450]);
plot_times = [0, 1, 3, 5, 10];
hold on;
for i = 1:length(plot_times)
    [~, t_idx] = min(abs(t_axis - plot_times(i)));
    plot(z_grid, M_matrix(:, t_idx), 'LineWidth', 2, 'DisplayName', [num2str(plot_times(i)) ' Seconds']);
end
title('Photolithography: PAC Concentration (M) Profile across Photoresist Depth');
xlabel('Depth into Photoresist (z, \mum)'); ylabel('Normalized PAC Concentration (M)');
ylim([-0.05 1.05]); legend('show', 'Location', 'southwest'); grid on;

% 4. ETCH PROFILE EVOLUTION (WET VS DRY RIE)
target_oxide_thick = 0.5;
mask_opening_width = 3.0;

figure(4);
set(gcf, 'Name', '2D Cross-Section Etch Profiles', 'Color', 'w', 'Position', [200, 200, 1100, 480]);

```

```

% Isotropic Wet Etch
subplot(1,2,1);
generate_etch_geometry_plot(target_oxide_thick, mask_opening_width, 'wet');
title('Wet Chemical Etching Profile (Isotropic, A = 0.05)');

% Anisotropic RIE Etch
subplot(1,2,2);
generate_etch_geometry_plot(target_oxide_thick, mask_opening_width, 'dry');
title('Reactive Ion Etching (RIE) Profile (Anisotropic, A = 0.92)');

% HELPER FUNCTIONS
|
% 1. Deal-Grove Kinetic Growth Engine
function [X] = deal_grove_model(T_celsius, t_hours, ambient, orientation)
    k = 8.617e-5;
    T = T_celsius + 273.15;

    if strcmp(ambient, 'dry')
        B0 = 772; Ea_B = 1.23; Xi = 0.02;
        if strcmp(orientation, '100'); BA0 = 3.71e6; Ea_BA = 2.00; else; BA0 = 6.23e6; Ea_BA = 2.00; end
    else
        B0 = 386; Ea_B = 0.78; Xi = 0;
        if strcmp(orientation, '100'); BA0 = 9.70e7; Ea_BA = 2.05; else; BA0 = 1.63e8; Ea_BA = 2.05; end
    end

    B = B0 * exp(-Ea_B / (k * T));
    B_over_A = BA0 * exp(-Ea_BA / (k * T));
    A = B / B_over_A;
    tau = (Xi^2 + A * Xi) / B;

    X = (-A + sqrt(A^2 + 4 * B * (t_hours + tau))) / 2;
end

```


% 2. Dill's Photolithography Exposure Engine

```
function [M, z, t] = dills_lithography_engine(thickness, t_max, I0, A, B, C)
    Nz = 60; Nt = 200;
    z = linspace(0, thickness, Nz)';
    t = linspace(0, t_max, Nt);
    dz = z(2) - z(1);
    dt = t(2) - t(1);

    M = ones(Nz, Nt);
    I = zeros(Nz, Nt);
    I(1, :) = I0;

    for j = 1:(Nt-1)
        for i = 2:Nz
            local_absorption = 0.5 * ((A * M(i-1, j) + B) + (A * M(i, j) + B));
            I(i, j) = I(i-1, j) * exp(-local_absorption * dz);
        end
        M(:, j+1) = M(:, j) .* exp(-I(:, j) * C * dt);
    end
end
```

% 3. Geometrical Etch Profile Plotter Engine

```
function generate_etch_geometry_plot(oxide_h, mask_w, mode)
    hold on;
    x_grid = linspace(-4, 4, 400);
    x_left = -mask_w / 2;
    x_right = mask_w / 2;

    % Silicon Substrate Base
    fill([-4 4 4 -4], [-0.5 -0.5 0 0], [0.7 0.7 0.7], 'EdgeColor', 'none', 'FaceAlpha', 0.5);
    text(-3.5, -0.25, 'Si Substrate', 'FontWeight', 'bold');

    % Left and Right Photoresist Masks
    fill([-4 x_left x_left -4], [oxide_h oxide_h oxide_h+0.4 oxide_h+0.4], [0.9 0.6 0.2], 'EdgeColor', 'k');
    fill([x_right 4 4 x_right], [oxide_h oxide_h oxide_h+0.4 oxide_h+0.4], [0.9 0.6 0.2], 'EdgeColor', 'k');
    text(-3.5, oxide_h+0.2, 'PR Mask', 'Color', [0.5 0.2 0]);
```

```

% Profile Coordinate Calculation
y_profile = zeros(size(x_grid));
for idx = 1:length(x_grid)
    cx = x_grid(idx);
    if strcmp(mode, 'wet')
        if cx < x_left
            dist = x_left - cx;
            if dist < oxide_h
                y_profile(idx) = oxide_h - sqrt(oxide_h^2 - dist^2);
            else
                y_profile(idx) = oxide_h;
            end
        elseif cx > x_right
            dist = cx - x_right;
            if dist < oxide_h
                y_profile(idx) = oxide_h - sqrt(oxide_h^2 - dist^2);
            else
                y_profile(idx) = oxide_h;
            end
        else
            y_profile(idx) = 0; % Cleared trench
        end
    else % 'dry' anisotropic
        if cx <= x_left || cx >= x_right
            y_profile(idx) = oxide_h;
        else
            if (cx > x_left && cx < x_left + 0.1)
                y_profile(idx) = (oxide_h / 0.1) * (cx - x_left);
            elseif (cx < x_right && cx > x_right - 0.05)
                y_profile(idx) = (oxide_h / 0.05) * (x_right - cx);
            else
                y_profile(idx) = 0;
            end
        end
    end
end

% Silicon Dioxide Layer
fill([-4 x_grid 4], [0 y_profile 0], [0.3 0.6 0.9], 'EdgeColor', 'b', 'FaceAlpha', 0.7);
text(0, oxide_h/2, 'Etched Trench Zone', 'HorizontalAlignment', 'center', 'Color', 'blue');

% Axis Formatting
xlabel('Wafer Position Axis (x, \mum)'); ylabel('Vertical Stack Height (y, \mum)');
xlim([-4 4]); ylim([-0.5 1.5]); grid on; axis equal;
end

```